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A Generative AI-Driven Systems Engineering Maturity and Cost Modeling Framework

Ray Madachy

Naval Postgraduate School

Ryan Bell, Ryan Longshore

Naval Information Warfare Center and Naval Postgraduate School

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Introduction

- ❑ Generative AI assistance is quickly transforming systems engineering processes
 - AI tools with large language models (LLMs) enable task automation, architecture generation, design, trade-off analysis, implementation, testing, and more.
 - Supports decision-making, risk management, and continuous process improvement.
 - Drastic reductions in effort when applied effectively
- ❑ Addressing new challenges in disrupted cost models for systems and software engineering
- ❑ Why this framework?
 - Provides a structured approach for organizations to integrate AI effectively and optimize engineering processes.
 - Harmonizes AI adoption maturity and costs models
 - Aligns AI usage with quantitative cost impacts to better inform decision-makers.

Framework Overview

- ❑ Harmonized maturity and cost models.
 - Maturity model provides a roadmap for AI adoption, identifying key practices with five progressive levels.
 - Parametric cost model quantifies the cost impact of AI usage on systems engineering projects.
 - Underlying ISO/IEC/IEEE15288 [6] harmonization of systems and software engineering activities.
- ❑ Introduction of *AI Usage* factor for cost modeling using Constructive Cost Model (COCOMO) [8] and Constructive Systems Engineering (COSYSMO) [7] framework.
- ❑ Empirical data collection and process for cost model calibration.
- ❑ The maturity levels and cost factor rating scale for *AI Usage* also serve as a foundation for collecting empirical data aligned with practices and activities in the ISO/IEC/IEEE15288 lifecycle standard [6].
- ❑ This iterative feedback with cost modeling refines predictive accuracy and facilitates continuous improvement to optimize their AI adoption and integration strategies over time.

New COSYSMO Cost Factor

Constructive Systems Engineering Cost Model (COSYSMO)

System Size

	Easy	Nominal	Difficult
System Requirements	11	23	3
System Interfaces	2	6	5
Critical Algorithms	5	5	2
Operational_Scenarios	3	4	2

Cost Factors

Requirements Understanding	High	Documentation	Nominal	AI Usage	High
Architecture Understanding	Low	Diversity of Installations/Platforms	Nominal	Personnel Experience/Continuity	Nominal
Level of Service Requirements	Nominal	Recursive Levels in Design	Nominal	Process Capability	Nominal
Migration Complexity	Very High	Stakeholder Team Cohesion	High	Multisite Coordination	Nominal
Technology Risk	Nominal	Personnel/Team Capability	Nominal	Tool Support	Nominal

Labor Rate

Cost per Person-Month (Dollars)

Calculate

Data Collection and Analysis Methods

- ❑ Multi-project data collection in conjunction with other cost factors, which was used to develop COSYSMO [7] and COCOMO II [8].
- ❑ Controlled group experiments comparing activities performed with and without AI assistance.
 - Naval Information Warfare Center (NIWC) systems engineering activities.
 - Public sources, e.g., GitHub study [10].
- ❑ Small-scale empirical case studies in the classroom and performed by researchers.
- ❑ Delphi surveys for expert judgment.
- ❑ Bayesian approaches combining empirical project data and Delphi results.

Maturity Model Overview

- ❑ Organizations progress through levels by upgrading infrastructure, data management, workforce training, and governance. Key enablers include:
 - Robust data management and access infrastructure are critical for supporting generative AI tools, which rely on large datasets for training and inference.
 - Engineers must be trained to effectively use generative AI tools. This includes not only technical skills but also an understanding of AI limitations and how to interpret AI-generated outputs.
 - Successful adoption requires rethinking engineering workflows to fully integrate AI-generated insights, ensuring that AI contributes meaningfully to systems engineering goals.
 - Clear policies must be established regarding the use of AI in engineering, including issues of accountability, transparency, and bias mitigation.
- ❑ Maturity Levels
 - Level 1: Initial - Minimal AI adoption, mostly traditional engineering workflows.
 - Level 2: Developing - AI usage in select tasks like documentation, basic automation.
 - Level 3: Competent - AI regularly used in design, modeling, and analysis.
 - Level 4: Advanced - AI plays a strategic role in decision-making and optimization.
 - Level 5: Optimized - AI is fully integrated, automating most engineering workflows.

Maturity Model Key Practices

- ❑ **Investment in Hardware and Infrastructure** addresses the technological resources required for AI adoption, ranging from basic infrastructure to highly scalable, robust systems.
- ❑ **Knowledge Integration and Data Management** highlights how AI systems evolve from limited context awareness to fully indexed and dynamically managed knowledge bases.
- ❑ **LLM Customization and Fine-Tuning** describes the depth of AI model customization, from using a generic model to extensive fine-tuning with specialized datasets.
- ❑ **LLM Retraining** indicates the frequency and adaptiveness of model updates, evolving from minimal retraining to real-time adaptive learning with Reinforcement Learning from Human Feedback (RLHF).
- ❑ **Personnel Training** captures the progressive development of workforce competencies to improve AI-assisted engineering processes. Without adequate training, organizations risk misuse, over-reliance, or underutilization of AI tools, leading to inefficiencies or defective engineering outputs.

Maturity Levels

Practice	Level 1: Initial	Level 2: Developing	Level 3: Competent	Level 4: Advanced	Level 5: Optimized
Investment in Hardware and Infrastructure	Minimal, basic cloud subscription or server for API calls.	Low to moderate, added storage and processing for few-shot or limited RAG/fine-tuning.	Moderate, scalable storage and processing for an evolving knowledge base and RAG integration.	High, GPU-based infrastructure for extensive fine-tuning and large data processing.	Very high, robust infrastructure for multi-agent orchestration, real-time updates, and continuous learning.
Knowledge Integration and Data Management	Minimal, relying solely on model's inherent knowledge.	Basic, limited context-awareness using few-shot or small, static RAG knowledge base.	Moderate, using a fully indexed knowledge base to improve context-aware responses.	High, large, domain-specific knowledge base updated regularly for accuracy.	Real-time, cooperative agents with dynamic retrieval, high context sensitivity, and advanced data management.
LLM Customization and Fine-Tuning	None, using a generic pre-trained model without customization.	Minimal, few-shot learning or minor fine-tuning with small datasets.	Moderate, fine-tuning for specific tasks or domains with RAG integration.	High, extensive fine-tuning with proprietary or domain-specific data for customization.	Extensive, with specialized agents and continual fine-tuning for adaptability.
LLM Retraining	None, no retraining—uses only pre-trained model.	Basic periodic retraining with small data samples as needed.	Scheduled fine-tuning sessions based on new domain-specific data.	Continuous domain-specific retraining at intervals or after major data changes.	Real-time or adaptive retraining using RLHF and continuous updates with interaction.
Personnel Training	No formal training on AI tools; usage is ad hoc and based on personal initiative.	Limited training programs focused on basic AI tool usage and guidelines.	Standardized training on AI-assisted workflows, with emphasis on integrating AI into engineering tasks.	Advanced training programs incorporating best practices for AI-human collaboration and workflow optimization.	Continuous learning culture with AI literacy deeply embedded in engineering practices, with mentorship and adaptive training based on real-time AI evolution.

Project Cost Factor Ratings

Very Low	Low	Nominal	High	Very High
Minimal to no AI assistance.	Moderate AI usage, typically for summarization, basic information retrieval, and model templating.	Regular use of AI tools for various tasks like MBSE help, design insights, or testing assistance.	Frequent and strategic use of AI assistance.	AI tools are deeply ingrained in most activities. They are crucial for decision making, problem solving, and automating tasks.
Development relies on traditional methods and tools.			AI tools play a central role across the lifecycle from architecting, to design, implementation, and V&V.	
AI tools may be present but are rarely, if ever, consulted.	AI tools are not deeply integrated into the development workflow.	AI tools are a recognized part of the toolkit but aren't central to development		The development process is designed around maximizing AI tool benefits.

Example Investment Analysis

- ❑ A systems engineering organization of 100 people is currently at maturity Level 2 (Developing) and wants to advance to Level 3 (Competent). They develop 1500 Equivalent Nominal System Requirements per year on projects at \$20,000/Person-Month.
- ❑ It is desired to estimate the annual cost savings after AI adoption and break-even point when the investment will pay for itself.
- ❑ Investment costs to achieve Level 3 per maturity level practices:

Expense	Cost
All systems engineers take generative AI training	\$300K
High performance computers purchased for LLM training	\$150K
LLM team will retrain organizational models	\$180K
Total Investment	\$630K

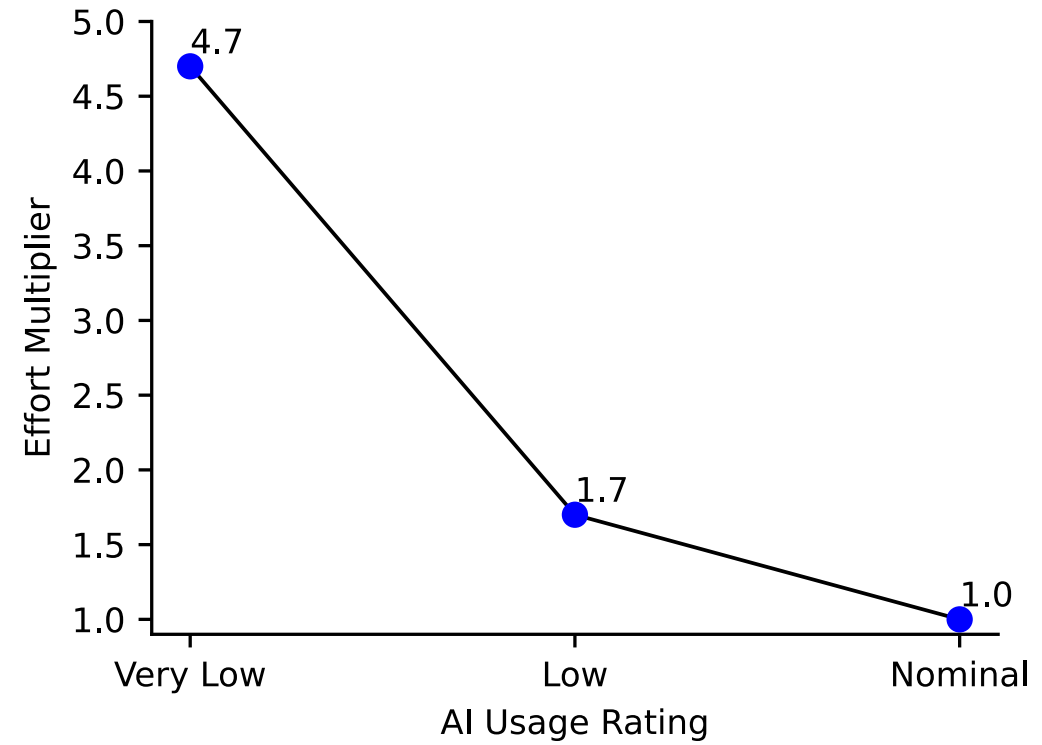
- ❑ After adoption, their project practices will improve from the *AI Usage* rating from Low to Nominal.
- ❑ When is the break-even point?

COSYSMO Effort Equation and Provisional *AI Usage* Effort Multipliers

$$Effort = A * Size^B * \prod_{i=1}^N EM_i$$

Where

- *Effort* is in Person-Months (PM)
- *A* is a constant derived from historical project data
- *Size* is a sum of weighted system requirements, interfaces, algorithms and scenarios
- *B* is an exponent for the diseconomy of scale
- EM_i is an effort multiplier for the i^{th} cost driver. The geometric product of N multipliers is an overall Effort Adjustment Factor (*EAF*) to the nominal effort.



Investment Analysis Results

❑ As-Is (AI Usage = Low)

$$Effort = A * Size^B * EAF = .054 * 1500^{1.06} * 1.7 = 213.6 \text{ PM}$$

$$\text{Annual Cost} = \$20,000/\text{PM} * 213.6 \text{ PM} = \$4.27\text{M}$$

❑ To-Be (AI Usage = Nominal)

$$Effort = A * Size^B * EAF = .054 * 1500^{1.06} * 1.0 = 125.6 \text{ PM}$$

$$\text{Annual Cost} = \$20,000/\text{PM} * 125.6 \text{ PM} = \$2.51\text{M}$$

❑ From the above, the annual savings and break-even point can be calculated:

$$\text{Annual Savings} = \$4.27\text{M} - \$2.51\text{M} = \$1.76\text{M}$$

$$\text{Break Even} = \text{Investment} / \text{Annual Savings} = \$0.63\text{M} / \$1.76\text{M} = .36 \text{ years}$$

❑ The investment will pay for itself in slightly over 4 months of adoption on systems engineering projects.

❑ Thus, it would be a worthwhile investment, and the ROI would be a substantial multiplier after a few years.

Conclusions and Future Work

- ❑ AI is transforming systems engineering, enabling efficiency, creativity, and better decision-making.
- ❑ Organizations must invest in training, infrastructure, and governance to optimize systems engineering processes with generative AI.
- ❑ The integrated maturity and cost modeling framework offers a structured pathway for organizations.
 - Serves as a roadmap for AI adoption, helping organizations understand where they currently stand and what steps are needed to advance.
 - Provides a basis for benchmarking progress and identifying best practices, fostering a culture of continuous improvement.
 - Cost model integration enables better informed cost and schedule estimates for projects using generative AI.
 - Supports holistic decision-making that considering technical capabilities, personnel skills, and and cost impacts.
- ❑ Future Work
 - Ongoing empirical calibration of cost models with real-world AI-assisted projects.
 - Refine of the *AI Usage* cost factor based on ongoing industry feedback.
 - Develop of open-source AI cost modeling tools.
 - Further analyze AI impacts across the lifecycle aligning artifacts and effort data with ISO/IEC/IEEE 15288.

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